Center-Wise Feature Consistency Learning for Long-Tailed Remote Sensing Object Recognition

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***Abstract*— Long-tailed distribution of remote sensing data generally limits the object recognition performance of deep neural networks. We notice that too many samples from head class will induce the neural network to learn features of tail class samples being biased toward the head. To solve this, we propose a novel center-wise feature consistency learning (CFCL) mechanism for long-tailed remote sensing object recognition. First, we implement a head-tail center feature generation procedure that builds two teacher models to extract the knowledge from the head class and tail class samples, respectively, so as to avoid the extracted tail class features being affected by the head classes. Second, a CFCL strategy is introduced, which distills the central feature of each class to a student model, thereby making the classification boundaries more prominent. Especially, the central feature is estimated by referring to the features which are correctly classified by the teacher models, thus the inaccurate knowledge is abandoned. Extensive experiments on widely adopted remote sensing recognition datasets including FGSC-23, DIOR, xView, and HRSC2016 demonstrate that our method achieves superior performance compared to the state-of-the-art approaches. Code and data are available at: https://github.com/wdzhao123/CWFC.**

***Index Terms*— Center-wise feature consistency learning (CFCL), head-tail center feature generation, long-tailed distribution, remote sensing object recognition.**

I. Introduction

DEEP convolutional neural networks (CNNs) have achieved remarkable progress in object recognition and detection tasks during the past decade. Simultaneously, challenging benchmarks such as ImageNet [1] and COCO [2] were successively proposed and widely adopted to evaluate the performance of related approaches due to their massive scale and relatively ideal sample distribution. Despite containing many different classes in total, the number of samples from each class is typically evenly distributed, meaning that even rare classes have sufficient samples for training. Yet, in the field of the remote sensing, the sample distributions among classes are usually uneven because of the extremely imbalanced distribution of actual objects in real world. For example, the aircraft carrier class often contains limited image samples, while the fishing boat class always cover a large number of ones. In other words, that kind of distribution presents a long-tailed characteristic (i.e., a few of head classes occupy the majority of samples, while the remaining tail classes only involve a small part). Unfortunately, learning directly from long-tailed data will probably causing a significant performance degradation when using a deep learning model, since plenty of head class samples will force the model to learn features of the tail class samples being biased toward the head. Therefore, how to improve the learning stability of neural networks on long-tailed remote sensing dataset remains a valuable topic.

Classical object recognition models, e.g., VGG [3] and ResNet [4], are widely utilized due to their simplicity and efficiency. Besides, advanced methods improve network designs [5], [6], [7], [8], [9] and optimize training strategies [10], [11], [12], [13], [14]. For example, [5] proposes a mathematical framework to design high-performance CNN models, and the effectiveness of a CNN network can be analytically formulated by its structural parameters. Shen et al. [9] proposes Equiangular Basis Vectors to replace traditional fully connected layers, achieving a fixed predefined embedding for all the semantic categories. Zhang et al. [14] designs an abstraction-based training method for robust classification models, where images are perturbed and mapped into intervals before feeding into neural networks for training. However, these methods ignore unevenly distributed classes. As for the long-tailed condition, existing methods usually implement class rebalancing strategies to mitigate the deterioration, such as resampling training data [15], [16], [17], [18], [19] and designing cost-sensitive reweighting loss functions [20], [21], [22], [23], [24], [25], [26]. The resampling method adjusts the unbalanced data distribution by highly sampling the tail classes. While reweighting methods assign higher weights to tail class samples when calculating losses. Generally, these methods can reduce the dominant role of the head classes during the training process, so as to produce accurate classification decision boundaries. Yet, modifying the original data distribution increases the possibility of overfitting to the tail classes.

A very recent trend is training a multiexpert network by organizing long-tailed data into groups, then combining the results from experts with various procedures [12], [27], [28],

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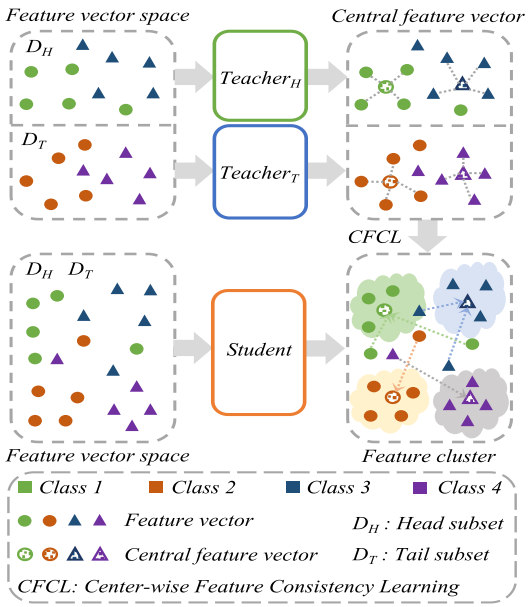


Fig. 1. Key idea of our approach, which can be divided into two stages: head-tail center feature generation and CFCL. Head subset D_H and tail subset D_T are used to train the corresponding teacher models Teacher_H and Teacher_T, respectively. Thus, the tail-sample feature learning will not be affected by the head samples. After that, the central feature of each class is estimated and transferred to the student. The divide-and-conquer feature learning mechanism for D_H and D_T prevents the student model from being biased toward the head classes.

[29], [30], [31], [32]. These methods usually include two training stages. The first stage is to train multiple teacher models (experts), and the second stage is to transfer the abilities of teacher models to a single student model. Many creative approaches have been proposed to conduct the transfer work, e.g., simple mean strategy [27], [33], [34], [35], complex knowledge distillation process [36], [37], [38], [39], [40], and soft label generation [41], [42]. Similar research routes show great promises to relieve the long-tailed problem.

The main difficulty of long-tailed remote sensing object recognition is that the learning of tail classes features can be affected by the head classes, resulting in the inaccurate classification on tail classes. In this article, we propose a center-wise feature consistency learning (CFCL) to address this issue. As shown in Fig. 1, our framework can be divided into two stages, including head-tail center feature generation and CFCL. During the first stage, we divide the original dataset D into two subsets (i.e., head subset D_H and tail subset D_T) based on the number of samples in each class. D_H and D_T are, respectively, fed into their corresponding teacher models T_H and T_T for training. Since using two teacher models T_H and T_T can avoid the feature space being dominated and biased by the head classes, this procedure can eliminate the long-tailed problem. Then, we select the features which are correctly classified by T_H and T_T to estimate the central features v_{c_T}. In this way, the student model S will only receive the correct information from T_H and T_T. During the second stage, we transfer the knowledge to the student model S by making the feature extracted by S closer to its corresponding central feature v_{c_T}. Therefore, features of each class extracted by S will be more compact, which will improve the performance of the long-tailed remote sensing object recognition task. The contributions of this article can be summarized as follows.

1) We spot a point that greatly influences the long-tailed remote sensing object recognition, i.e., the imbalanced data distribution will force the model to have a bias toward the majority classes, thereby neglecting the minority classes.
2) By dividing the training set into multiple subsets and training their corresponding teacher models, we are able to minimize the interactions between head and tail classes.
3) Center-wise feature distillation can suppress the noise generated by teachers, which, as a result, improves the efficiency of transfer learning.

II. Related Work

A. Long-Tailed Object Recognition

Existing methods solve the long-tailed object recognition through either resampling [15], [16], [17], [18], [19], [43], [44], [45], [46], [47] or cost-sensitive reweighting [13], [20], [21], [22], [23], [24], [25], [48], [49], [50]. Resampling methods can be divided into two types: 1) oversampling samples the tail classes repeatedly [15], [16], [17], [43], [44] and 2) undersampling clamps down the head classes [17], [19], [45], [46], [47]. For example, Ando and Huang [43] propose deep oversampling that extends the synthetic oversampling method to the deep feature space. Yu et al. [47] cope with tail classes and head classes in a dual-path manner to boost the performance on tail classes. Cost-sensitive reweighting methods usually assign large weights to the tail classes and relatively small weights to the head classes. Thus, the loss function pays more attention to the tail classes, thereby strengthening the optimization of the tail classes. For instance, Alshammari et al. [20] use a cross-entropy loss to learn features as well as a class balance loss to learn classifiers. Wang et al. [25] design a class-aware angle margin loss, where the difference between two categories is proportional to the weight specifications of their classifiers.

However, the oversampling method repeatedly samples the tail data, which may result in overfitting of the tail data. Undersampling ignores part of the head data (i.e., part of the head data may not be sampled for training). In addition, reweighting assigns weights to different classes, which is sensitive to the selection of weighting factors.

B. Multiexpert Learning

Recently, the emergence of multiexpert networks [27], [28], [29], [30], [33], [36], [37], [45], [51], [52] brings a brand-new view. Zhou et al. [27] propose a unified bilateral branch network to handle both presentation learning and classifier learning, where each branch performs its own responsibilities separately and is equipped with a cumulative learning strategy. By doing so, the model can learn the general patterns first and then gradually focus on the tail data. Wang et al. [33] propose a hybrid network structure consisting of supervised contrast loss to learn image representation and cross entropy loss to learn classifiers, which can embody better features and

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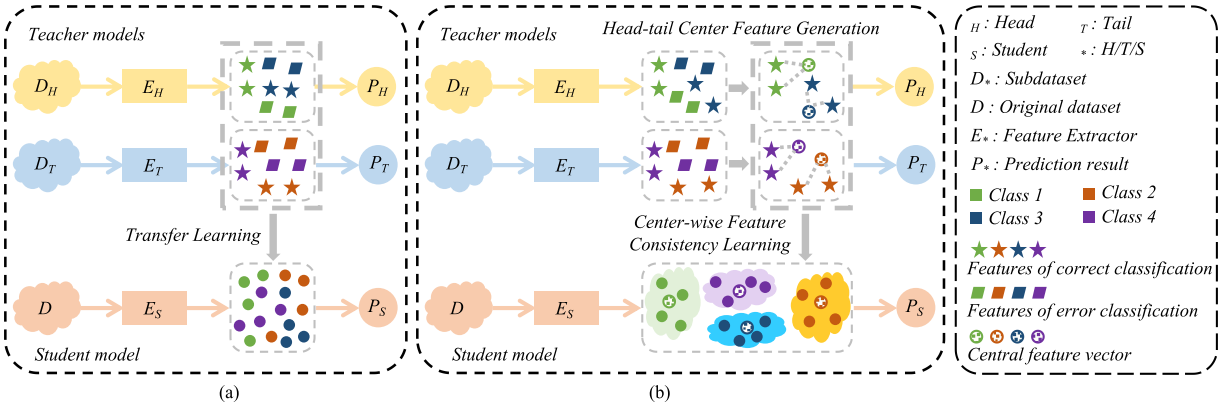


Fig. 2. Principle comparison between (a) previous multiexpert distillation framework and (b) our CFCL framework. (a) Transfers the knowledge of the teacher models to the student model without restriction, and thus the wrong knowledge may be transferred. By contrast, (b) estimates a central feature for each class as knowledge to be transferred, which effectively lower the impact of noise from teacher models.

produce better classifiers. He et al. [37] solve the long-tailed visual recognition problem from the perspective of knowledge distillation and propose an extraction virtual instance method by using the prediction of the teacher model as a virtual sample.

Inspired by multiexpert learning, we repartition the original training set into head and tail subsets according to its data distribution. Then, we use the head and tail subsets to train two experts, respectively, which can avoid the learning of tail class features being affected by the head classes. After that, we implement a CFCL to obtain the student model with better performance for the long-tailed remote sensing object recognition task.

C. Knowledge Distillation

Knowledge distillation aims to transfer the knowledge learned from a large model to a lightweight one, so that it can obtain the advanced ability from its teacher with less cost. Knowledge distillation have been widely adopted to solve long-tailed problem. Parisot et al. [39] leverage the fact that class prototypes and learned cosine classifiers provide two complementary representations of class cluster centers in feature space, and use an attention mechanism to recompose learned classifier features from common classes to obtain higher quality rare-class representations. Li et al. [40] introduce a self-distillation framework for long-tailed recognition and propose a distillation label generation module based on self-supervision. Ju et al. [42] propose class subset learning by dividing the long-tailed data into multiple subsets according to prior knowledge, such as regions and phenotype information. Zhao et al. [53] propose a weakly correlated distillation method that selects the weakly correlated features from teacher models to teach the student.

Unfortunately, if the knowledge of the teacher models is transferred to the student model without proper restrictions, the student may receive an incorrect signal. Therefore, we introduce the CFCL that estimate the central feature of each class to abandon the inaccurate knowledge. Through such a strategy, the student model will only receive the right knowledge to improve its recognition ability.

III. Proposed Method

A. Motivation and Framework

Multiexpert structure have been developed to handle the long-tailed problem for a long period of time [27], [28], [29], [30], [33], [36], [37], [45], [51], [52]. Though specific methods vary, they all follow similar principles as follows. Given a dataset D with long-tailed distribution that consists of total C classes, K thresholds are set to divide D into $K+1$ subsets according to the number of samples per class. For simplicity, we only employ two subsets D_H and D_T to train teacher models T_H and T_T, respectively. Next, the knowledge learned from the teacher models is transferred to the student model as shown in Fig. 2(a). Yet, the inaccurate knowledge from T_H and T_T may have negative impacts on the distillation procedure, which will greatly affect the ability of S.

In contrast, we overcome this challenge by counting the features of the samples correctly classified by T_H and T_T and calculating a corresponding central feature v_{c_T} for each class. As a result, the central features v_{c_T} are used to teach the student model S with precise knowledge. Specifically, we build a CFCL framework as shown in Fig. 2(b). Two stages are introduced in our CFCL structure: head-tail center feature generation and CFCL.

B. Head-Tail Center Feature Generation

Excessive head samples will lead to a fitting bias toward the head, thereby lower the recognition accuracy of tail classes. We employ two subsets and train their individual teacher model to prevent the impact from the head classes. Furthermore, we estimate the central feature of each class to suppress the inaccurate knowledge as shown in Fig. 3.

Specifically, we use the head subset D_H and tail subset D_T to train their corresponding teacher models T_H and T_T with loss functions,

$$L_H = -\frac{1}{N_H}\sum_{x_H \in D_H} P(x_H)\log(f_H(x_H)) \tag{1}$$

$$L_T = -\frac{1}{N_T}\sum_{x_T \in D_T} P(x_T)\log(f_T(x_T)) \tag{2}$$

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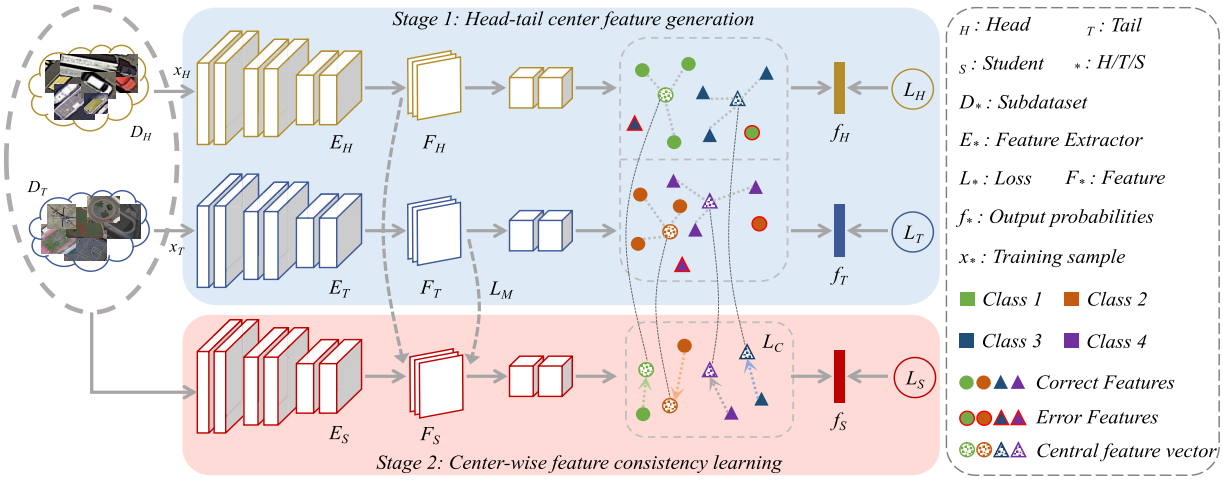


Fig. 3. Structure illustration of the proposed method. Two stages including head-tail center feature generation and CFCL are implemented. In stage 1, D_H and D_T are used to train teacher models T_H and T_T, respectively. Then, the central feature vector of each class is estimated according to the samples correctly classified by the teacher. In stage 2, the central feature vectors and the spatial features F_H and F_T are distilled to the student S that makes features of each class more compact.

where x_H and x_T stand for training samples from different subsets. N_H and N_T are the total number of samples in two subsets. P_{x_H} and P_{x_T} represent the ground truth of x_H and x_T. $f_H(x_H)$ and $f_T(x_T)$ indicate the output probability of T_H and T_T.

Now, we have T_H and T_T with great feature extraction abilities, which also means the precise feature expression vectors $v(x_i), i = H/T$ can be obtained by feature extractors E_H and E_T (see Fig. 3). Since the teacher models inevitable suffer from misclassification, only samples that are correctly recognized by T_H and T_T will be selected to participate in the estimation of central feature vectors v_c, where c represents the category. The estimation formula can be expressed as

$$v_{c_T} = \frac{1}{n_{c_T}} \sum_{x_{c_T} \in D_R} v\left(x_{c_T}\right) \tag{3}$$

where D_R is the set of correctly classified samples. x_{c_T} stands for the sample from class c in D_R. n_{c_T} is the total number of samples in class c. And v_{c_T} represents the central feature vector of class c.

C. Center-Wise Feature Consistency Learning

The central feature vectors v_{c_T} provide comprehensive descriptions of the knowledge from teacher models T_H and T_T, which can also suppress the inaccurate noise. Thus, we need a relevant distillation mechanism to leverage those picked up knowledge for student model training. As shown in Fig. 3, we introduce a CFCL to achieve this goal. Specifically, the feature vector $v_S(x_c)$ extracted by student model S will be pushed toward its corresponding v_{c_T} by minimizing their Euclidean distance in embedding space. The loss function is defined as

$$L_C = \frac{1}{N_B} \sum_{x_c \in D_c} \left(v_S(x_c) - v_{c_T}\right)^2 \tag{4}$$

where N_B stands for batch size. D_c means set of samples with class c. And $v_S(x_c)$ is the feature vector generated by the feature extractor E_S of student model S.

By doing so, the student model is forced to extract features that are closer to the noise-suppressed central features from its teacher, thus forms a compact embedding cluster for different classes and provides prominent classification boundaries for the classifier. In addition, to guarantee the fundamental recognition ability of S, we build its classification loss as follows:

$$L_S = -\frac{1}{N_B} \sum_{x_i \in D_T} P(x_i) \log(f_S(x_i)) \tag{5}$$

where x_i stands for the training sample in D. N_B is the number of training samples in D. $P(x_i)$ represents the ground truth label of x_i. And $f_S(x_i)$ indicates the output probability of S.

Moreover, we act a basic feature distillation procedure on spatial features F_H and F_T from teacher model E_H and E_T. Specifically, giving a training image $x_i, i = H/T$ sampled from $D = D_H, D_T$, we distill the feature representation $F_i(x_i)$ from its corresponding teacher T_i to S with following loss:

$$L_M = \frac{1}{N_B} \sum_{x_i \in D} (F_S(x_i) - F_i(x_i))^2 \tag{6}$$

where $F_S(x_i)$ is the feature generated by S from sample x_i. N_B means batch size.

Overall, the total loss L_{TS} for the whole CFCL structure is defined as

$$L_{TS} = L_S + L_C + L_M. \tag{7}$$

D. Optimizing

1) Structure: We employ a ResNet-50 [4] as our backbone, which consists of four residual blocks in total. A global mean pooling layer is supplemented after the fourth residual block in order to better estimate the central feature vector used in our framework. A simple fully connected operation is performed as classifier to facilitate recognition task. Not only teacher models T_H, T_T but also student model S share the same network structure.

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TABLE I
STATISTICAL DETAILS OF FGSC-23 DATASET

subset	Head							Tail															
Class	Non ship	Destoryer	Frigate	Cruiser	Submarine	Auxiliary ship	Bulk carrier	Aircraft carrier	Landing craft	Amphibious transport dock	Tarawa class amphibious assult ship	Amphibious assult ship	Command ship	Medical ship	Combat boat	Container ship	Car carrier	Hover craft	Oil tanker	Fishing boat	Passenger ship	Liquefied gas ship	Barge
Identification	SC0	SC1	SC2	SC3	SC4	SC5	SC6	SC7	SC8	SC9	SC10	SC11	SC12	SC13	SC14	SC15	SC16	SC17	SC18	SC19	SC20	SC21	SC22
Train Number	387	434	236	234	190	180	274	132	86	72	70	123	71	17	114	81	58	96	132	82	70	74	43
Test Number	97	108	59	59	48	45	69	34	22	18	18	31	18	10	29	20	14	24	33	20	18	20	11

TABLE II
STATISTICAL DETAILS OF DIOR DATASET

subset	Head								Tail											
Class	Airplane	Baseballfield	Harbor	Ship	Storagetank	Tenniscourt	Vehicle	Windmill	Airport	Basketballcourt	Bridge	Chimney	Dam	Expressway-Service-area	Expressway-toll-station	Golffield	Groundtrackfield	Overpass	Stadium	Trainstation
Identification	TA0	TA1	TA2	TA3	TA4	TA5	TA6	TA7	TA8	TA9	TA10	TA11	TA12	TA13	TA14	TA15	TA16	TA17	TA18	TA19
Train Number	1892	2384	2369	27290	3053	4906	13713	2365	661	1076	1363	649	512	1080	610	511	1158	1336	595	502
Test Number	8212	3433	3140	35106	23361	7360	26654	2998	666	2149	2604	1032	537	1085	688	575	1880	1778	673	509

2) Training: The whole learning process consists of two stages as shown in Algorithm 1. During the first stage, we divide the original dataset D into two subsets D_H and D_T for training two teacher models T_H and T_T, respectively. Then, samples correctly recognized by T_H and T_T are collected and their features are provided to estimate the central feature vector v_{c_T}. For the second stage, the features generated by student S are forced closer to their corresponding central feature vectors v_{c_T}. Meanwhile, spatial features of T_H and T_T are transferred to student S as basic information guidance.

Algorithm 1 Optimization process of our framework

Input: Head and tail subset D_H and D_T and original dataset D.
Stage 1: Training teacher models T_H and T_T with D_H and D_T, respectively, and generating head-tail center feature.
1. Optimizing T_H and T_T using (1) and (2).
2. Generating center feature using (3) by the well-trained T_H and T_T.
Stage 2: Center-wise feature consistency learning of student model S with D.
3. Optimizing S using (7).
Output: Parameters of S.

IV. EXPERIMENTS

A. Settings

1) Datasets: Four widely accepted long-tailed remote sensing datasets including FGSC-23 [54], DIOR [55], xView [56], and HRSC2016 [57] are adopted. In FGSC-23, 3256 images are used for training and 825 images are used for testing. The number of training samples for each class varied from 17 to 434 (statistical details are shown in Table I). In DIOR, 68 025 training samples and 124 440 testing samples are adopted. The number of training samples for each class ranges from 502 to 27 290 (see Table II for statistical details). In xView, we employ samples from 51 classes including 69 206 for training and 29 626 for testing (see Table III for details). HRSC2016 is limited in terms of the data size, meaning that only 1172 samples can be used to train a classification model (statistical details are shown in Table IV). Meanwhile, the sample distribution of different classes also remains imbalanced. Training samples from each class can be ranged from as low as 11 to up to 276. Twelve out of 18 classes have less than 40 samples. All of these pose a greater challenge to the generalization of classification approaches comparing with other datasets.

Under all data settings, the inputs are resized to 224×224 before further processing. Taking the number of samples into account, we utilize a two subsets separation (i.e., head subset and tail subset) for all datasets. The threshold is set to 150 for FGSC-23, 1500 for DIOR, 1000 for xView, and 50 for HRSC2016.

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TABLE III
STATISTICAL DETAILS OF XVIEW DATASET

Subset	Head																Middle									
Class	Building	Small Car	Truck	Bus	Cargo Truck	Vehicle Lot	Trailer	Truck w/Box	Utility Truck	Passenger Vehicle	Shipping Container Lot	Cargo Car	Storage Tank	Passenger Car	Shipping Container	Motorboat	Dump Truck	Construction Site	Shed	Pickup Truck	Damaged Building	Facility	Truck w/Flatbed	Truck Tractor	Excavator	Fishing Vessel
Train Number	9530	8888	8533	4882	4128	2986	2873	2558	2544	2068	1665	1296	1199	1142	1113	1025	951	854	834	773	768	667	628	604	593	560
Test Number	4084	3809	3656	2092	1769	1279	1230	1095	1090	886	713	555	513	489	476	438	407	365	357	330	329	285	269	258	253	239

Subset	Tail																									
Class	Cargo Plane	Hut/Tent	Sailboat	Maritime Vessel	Front Loader/Bulldozer	Yacht	Pylon	Small Aircraft	Mobile Crane	Haul Truck	Container Ship	Cement Mixer	Tugboat	Aircraft Hangar	Container Crane	Barge	Crane Truck	Tower Crane	Truck w/Liquid	Flat Car	Helipad	Locomotive	Tank Car	Tower	Oil Tanker	Total
Train Number	503	499	493	483	441	305	262	258	237	237	2v33	201	159	147	136	135	122	110	105	88	88	87	86	60	59	69206
Test Number	215	213	210	206	189	135	112	110	101	101	99	86	68	62	58	57	52	46	44	37	37	37	36	25	24	29626

TABLE IV
STATISTICAL DETAILS OF HRSC2016 DATASET

subset	Head						Tail											
Class	Ship	Arleigh Burke class destroyers	Cargo ship	Ticonderoga class cruiser	Perry class frigate	Tarawa-class amphibious assault ship	Whidbey Island class landing craft	Hovercraft	Austen class amphibious transport dock	Nimitz class aircraft carrier	Oxo	Submarine	Sanantonio class amphibious transport dock	Transport ship	Container ship	Enterprise class aircraft carrier	Medical ship	Midway-class aircraft carrier
Identification	HR0	HR1	HR2	HR3	HR4	HR5	HR6	HR7	HR8	HR9	HR10	HR11	HR12	HR13	HR14	HR15	HR16	HR17
Train Number	276	183	122	106	100	66	46	39	38	36	36	26	21	21	19	15	11	11
Test Number	167	124	88	94	111	38	41	44	32	23	33	36	17	13	15	10	10	8

TABLE V
TOP1 ACCURACY (%) OF OUR METHOD AND THE BASELINE, HEAD TEACHER MODEL, AND TAIL TEACHER MODEL. AVE_ACC IS THE AVERAGE ACCURACY

		Head							Tail															
	Ave_acc	SC0	SC1	SC2	SC3	SC4	SC5	SC6	SC7	SC8	SC9	SC10	SC11	SC12	SC13	SC14	SC15	SC16	SC17	SC18	SC19	SC20	SC21	SC22
Baseline	86.30	90.72	87.04	91.53	**91.53**	93.75	62.22	79.71	91.18	81.82	83.33	**99.99**	**96.77**	72.22	**99.99**	93.10	80.00	**99.99**	**99.99**	66.67	65.00	66.67	**99.99**	**99.99**
Head Teacher	-	89.69	88.89	**94.92**	86.44	**95.83**	**86.67**	82.61	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Tail Teacher	-	-	-	-	-	-	-	-	**94.12**	81.82	**88.89**	88.89	87.10	**88.89**	**99.99**	**99.99**	80.00	92.86	**99.99**	84.85	**85.00**	**77.78**	**99.99**	**99.99**
Ours	**88.85**	**91.75**	**93.52**	93.22	86.44	91.67	75.56	**85.51**	88.24	**95.45**	72.22	94.44	**96.77**	83.33	**99.99**	89.66	**85.00**	92.86	**99.99**	**90.91**	55.00	66.67	**99.99**	**99.99**

2) Implementation Details: The experiments are conducted on a single NVIDIA RTX 2080 Ti GPU with PyTorch framework. The momentum of SGD is set to 0.9 and the batch size is set to 8. The initial learning rate is 0.001 and reduce by ten times every 30 epochs. Both the teacher model and the student model are trained for 100 epochs.

B. Ablation Studies

1) Effectiveness of Head-Tail Center Feature Generation and CFCL: In Section III, we propose a head-tail center feature generation and a CFCL. The head-tail center feature generation uses the head and tail subsets to train two teacher models, respectively, so that the impact of head classes can

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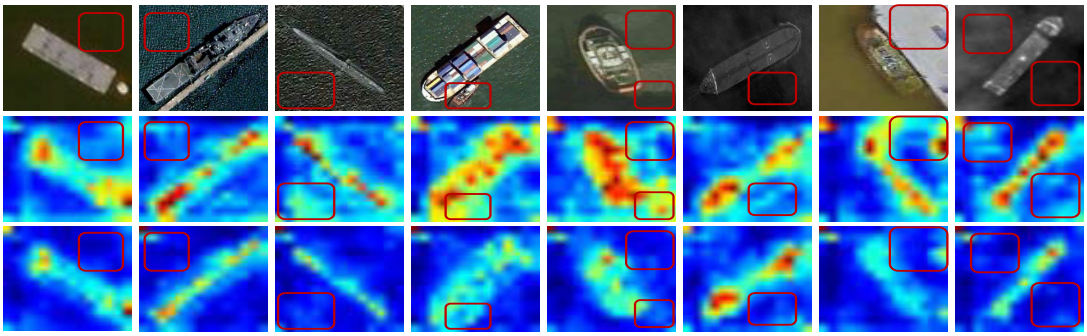


Fig. 4. Visualized feature maps of our approach and the baseline. The first to the third rows are original samples, feature maps of the baseline, and feature maps of our method. Feature maps are extracted from the third block of ResNet-50.

TABLE VI

EFFECTIVENESS OF SPATIAL FEATURE DISTILLATION BY COMPARING DIFFERENT-LEVEL FEATURE DISTILLATION. AVE_ACC IS THE AVERAGE ACCURACY

		Head							Tail															
	Ave_acc	SC0	SC1	SC2	SC3	SC4	SC5	SC6	SC7	SC8	SC9	SC10	SC11	SC12	SC13	SC14	SC15	SC16	SC17	SC18	SC19	SC20	SC21	SC22
1L-FD	86.42	89.69	91.67	**94.92**	**86.44**	**93.75**	60.00	79.71	91.18	72.73	72.22	**99.99**	**96.77**	**88.89**	**99.99**	**93.10**	**85.00**	**99.99**	**99.99**	78.79	45.00	61.11	**99.99**	**99.99**
2L-FD	87.15	89.69	92.59	**94.92**	**86.44**	91.67	73.33	75.36	**94.12**	90.91	**88.89**	**99.99**	90.32	77.78	**99.99**	**93.10**	**85.00**	92.86	95.83	78.79	**55.00**	55.56	**99.99**	**99.99**
3L-FD	**88.85**	**91.75**	**93.52**	93.22	**86.44**	91.67	**75.56**	**85.51**	88.24	**95.45**	72.22	94.44	**96.77**	83.33	**99.99**	89.66	**85.00**	92.86	**99.99**	**90.91**	**55.00**	**66.67**	**99.99**	**99.99**

TABLE VII

TOP1 ACCURACY (%) OF OUR METHOD TRAINING WITH ORIGINAL FGSC-23 DATASET AND HIGHER IMBALANCE ONE. AVE_ACC IS THE AVERAGE ACCURACY

		Head							Tail															
	Ave_acc	SC0	SC1	SC2	SC3	SC4	SC5	SC6	SC7	SC8	SC9	SC10	SC11	SC12	SC13	SC14	SC15	SC16	SC17	SC18	SC19	SC20	SC21	SC22
Original	**88.85**	**91.75**	**93.52**	93.22	86.44	91.67	**75.56**	**85.51**	88.24	**95.45**	**72.22**	**94.44**	**96.77**	83.33	**99.99**	89.66	**85.00**	**92.86**	**99.99**	**90.91**	55.00	**66.67**	**99.99**	**99.99**
Higher imbalance	83.88	88.66	89.81	**94.92**	**88.14**	**93.75**	64.44	82.61	**91.18**	86.36	66.67	38.89	**96.77**	**88.89**	80.00	**96.55**	**85.00**	35.71	**99.99**	84.85	**75.00**	22.22	**99.99**	54.55

be avoided. The CFCL estimates a central feature for each class, then distill those central features to the student model to improve the recognition accuracy. To verify their effectiveness, three control models including baseline trained with D, head teacher model T_H trained with D_H, and tail teacher model T_T trained with D_T are set up as shown in Table V. First, the head teacher model and the tail teacher model achieve 0.88% and 3.82% higher accuracy than the baseline, which illustrates that the divide-and-conquer feature learning prevents the tail features bias toward the head features. Second, our method outperforms the baseline model by 2.55%, since our approach relieves the problem that feature learning of tail samples is affected by head samples.

Visualized feature maps for our model and baseline model are depicted in Fig. 4. Our approach generates more consistent object features and less background noise compared to the baseline (see the red boxes).

2) Effectiveness of Spatial Feature Distillation: In Section III-C, we distill not only the class-level central features, but also the sample-level spatial features from T_H and T_T to student S. Since teacher models can extract features with better spatial semantic information after the first-stage training, we transfer those abilities to S to further improve the performance of student model. In order to verify the effectiveness of the spatial feature distillation, we conduct an ablation experiment testing distillations on different feature levels: first-level feature extracted from the first block output of ResNet-50 (1L-FD), second-level feature extracted from the second block output (2L-FD), and third-level feature extracted from the third block output (3L-FD). As shown in Table VI, 3L-FD achieves the best average accuracy of 88.85%. That is probably because deeper features contain stronger spatial semantic information of the objects.



Fig. 5. Example of low-resolution samples in DIOR dataset.

3) Study of Sample Imbalance Factor: We increase the imbalance factor between the head and tail by reducing the number of samples for some tail classes to less than ten in original FGSC-23 dataset (i.e., SC10, SC13, SC16, SC20, and SC22 contain nine, six, eight, nine, and seven samples, respectively). As shown in Table VII, when comparing to the results on original FGSC dataset, classes with less than ten samples encounter severe accuracy drop (e.g., SC10, SC16 and SC22). The reason may be that training samples for those tail

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Sample								
Ground truth	SC0	SC1	SC2	SC3	SC5	SC7	SC8	SC9
SADE	22.89%	1.13%	11.78%	7.33%	0.55%	1.62%	17.39%	4.49%
Ours	97.98%	99.96%	99.66%	99.99%	96.21%	93.83%	99.37%	92.50%
Sample								
Ground truth	SC10	SC13	SC14	SC16	SC18	SC19	SC20	SC21
SADE	5.78%	20.20%	31.11%	12.62%	12.32%	1.44%	43.34%	37.07%
Ours	99.87%	78.54%	99.66%	99.90%	52.69%	48.65%	93.45%	75.92%

Fig. 6. Qualitative comparison of our method and SADE method. The first row is the original samples, the second row is the ground truth, the third row is the accuracy recognized by SADE, and the fourth row is the accuracy recognized by our method.

TABLE VIII
RESULTS OF DIFFERENT NUMBERS OF TEACHER NETWORKS. AVE_ACC IS THE AVERAGE ACCURACY

Setting	Ave_acc
One teacher	58.64
Two teachers	61.35
Three teachers	61.51

classes are too few to afford a teacher model with acceptable feature extraction ability. Thus, both the central feature and the feature representation it provides will be of little help to the student model, or even mislead the student.

4) Number of Teacher Networks: We study the number of teacher networks when meeting the xView dataset which contains a large number of classes (see Table III for details). Specifically, three different subset division strategies are implemented, including one teacher (trained with one set of {head, middle, tail classes}), two teachers (trained with two subsets of {head, middle classes} and {tail class}), and three teachers (trained with three subsets of {head class}, {middle class} and {tail class}). As shown in Table VIII, two teachers improve the accuracy of student by 2.71% compared to one teacher. What is more, three teachers lead to a slightly better performance of 0.16% higher accuracy than two teachers. Comprehensively considering training cost and performance, we adopt two teachers setting for the experiments.

C. Comparison With the State of the Arts

We compare our proposed method with the state-of-the-art methods on FGSC-23, DIOR, and xView datasets, including SADE [58], TLC [35], RIDE [60], ResLT [61], BKD [59], and LAL [62]. To be noticed, RIDE, TLC, and ResLT use ResNet-50 as backbone, SADE uses ResNeXT-50 as backbone, and BKD and LAL use ResNet-32 as backbone. For fair comparisons, we utilize the original source codes provided by authors and fine-tune them with the corresponding training datasets.

1) Experimental Results on FGSC-23: Table IX reports the results of different methods on FGSC-23 dataset. We can see from the results that our method achieves the highest average accuracy of 88.85%, which is 18.06% higher than the second highest result 70.79% obtained by SADE. For individual classes, our method generally achieves better performance than the most advanced methods.

2) Experimental Results on DIOR: Table X shows the results of different methods on DIOR dataset. Our method also achieves a good effect of 91.89%. Our average accuracy outperforms the existing highest accuracy obtained by the method SADE by 3.49%. In addition, from the perspective of class-level accuracy, we achieve the best performance for 11 classes. Please notice that some single class does not reach the highest accuracy (e.g., TA1, TA3, TA4, TA10, TA11, TA12, and TA14). The reason may be that these classes have low-resolution samples, as shown in Fig. 5. However, our CFCL focuses on dealing with the long-tailed object recognition. Further study will integrate feature super-resolution strategy to relieve this. Combined with Tables IX and X, our method can improve the recognition effect of the tail classes, so as to relieve the problem of long-tailed remote sensing object recognition. In Fig. 6, we show the qualitative results between our method and the best existing approach SADE. Our method performs a better probability forecasting on object recognition.

3) Experimental Results on HRSC: Results of our CFCL and compared methods on HRSC are reported in Table XI. Our proposed CFCL maintains the superior performance and achieves the highest average accuracy of 78.98% among all testing approaches. As for class-level accuracy, our CFCL manages to take the lead in nine individual classes, including both majority classes and minority classes. We analyze that our method can reach a high performance mostly due to the mitigation of interactions between head and tail classes.

4) Experimental Results on xView: Results of our CFCL and compared methods on xView are reported in Table XII. Our CFCL obtains an average accuracy of 61.35%, which achieves the best performance than all the competitors. Especially, our CFCL outperforms the second best method TLC by 0.58%.

5) Computational Efficiency Analysis: Since our CFCL focuses on model training strategy that addresses the long-tailed sample distribution, the simple but efficient

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TABLE IX

TOP-1 ACCURACY (%) RESULTS OF OUR METHOD AND SIX EXISTING METHODS ON THE TEST SET OF FGSC-23. THE AVERAGE ACCURACY AND ACCURACY OF EACH CLASS ARE LISTED, AND THE BEST ACCURACY IS BOLDED. AVE_ACC IS THE AVERAGE ACCURACY

		Head							Tail															
Method	Ave_acc	SC0	SC1	SC2	SC3	SC4	SC5	SC6	SC7	SC8	SC9	SC10	SC11	SC12	SC13	SC14	SC15	SC16	SC17	SC18	SC19	SC20	SC21	SC22
SADE [58]	70.79	64.95	67.59	84.75	64.41	81.25	60.00	57.97	76.47	68.18	66.67	66.67	64.52	61.11	**99.99**	86.21	60.00	**92.86**	95.83	66.67	**65.00**	61.11	95.00	90.91
TLC [35]	44.12	46.39	21.30	28.81	23.73	43.75	28.89	10.14	64.71	22.73	61.11	33.33	45.16	61.11	**99.99**	86.21	60.00	85.71	91.67	57.58	60.00	**66.67**	**99.99**	**99.99**
BKD [63]	65.94	57.73	71.30	84.75	77.97	34.78	76.47	54.84	72.92	72.41	46.67	27.27	59.09	66.67	61.11	55.56	**99.99**	70.00	**99.99**	**99.99**	**65.00**	61.11	95.00	**99.99**
RIDE [60]	52.27	73.21	33.33	18.31	28.99	26.76	**82.81**	35.37	55.56	72.83	35.71	49.41	26.15	39.73	31.7 1	33.33	60.66	55.88	82.35	89.33	46.97	44.12	94.64	97.06
ResLT [61]	62.55	62.89	73.15	74.58	61.02	31.88	79.41	67.74	83.33	72.41	46.67	27.27	45.45	33.33	61.11	38.89	**99.99**	35.00	85.71	**99.99**	50.00	50.00	**99.99**	81.82
LAL [62]	53.45	75.26	72.22	42.37	18.64	47.83	64.71	58.06	68.75	79.31	42.22	21.21	27.27	27.78	27.78	5.56	80.00	50.00	42.86	91.67	40.00	44.44	45.00	**99.99**
Ours	**88.85**	**91.75**	**93.52**	**93.22**	**86.44**	**91.67**	75.56	**85.51**	**88.24**	**95.45**	**72.22**	**94.44**	**96.77**	**83.33**	**99.99**	**89.66**	85.00	**92.86**	**99.99**	90.91	55.00	**66.67**	**99.99**	**99.99**

TABLE X

TOP-1 ACCURACY (%) RESULTS OF OUR METHOD AND SIX EXISTING METHODS ON THE TEST SET OF DIOR. THE AVERAGE ACCURACY AND ACCURACY OF EACH CLASS ARE LISTED, AND THE BEST ACCURACY IS BOLDED. AVE_ACC IS THE AVERAGE ACCURACY

		Head								Tail											
Method	Ave_acc	TA0	TA1	TA2	TA3	TA4	TA5	TA6	TA7	TA8	TA9	TA10	TA11	TA12	TA13	TA14	TA15	TA16	TA17	TA18	TA19
SADE [58]	88.40	80.00	**90.50**	78.63	85.11	91.91	91.73	92.89	98.70	89.19	95.63	71.89	89.44	**87.34**	89.68	79.07	**98.43**	92.34	70.92	83.36	**95.48**
TLC [35]	82.70	68.47	78.68	71.75	83.78	87.81	89.02	84.54	94.13	76.58	88.60	46.70	81.78	73.18	83.41	60.03	94.26	89.51	65.92	72.21	84.48
BKD [63]	75.71	42.58	84.91	69.14	82.66	73.88	77.44	83.84	88.06	61.41	79.11	42.01	67.73	58.29	55.02	52.62	74.09	80.00	48.59	42.79	75.44
RIDE [60]	87.57	82.28	85.49	79.20	83.17	**92.37**	92.99	92.25	98.90	90.69	95.07	68.20	85.76	82.12	91.43	72.09	93.04	91.22	66.99	73.25	88.41
ResLT [61]	72.95	89.98	78.40	52.27	**99.00**	84.82	89.09	71.24	55.64	77.04	83.06	**75.00**	**90.41**	85.89	90.00	**85.81**	81.81	72.27	74.12	72.17	85.07
LAL [62]	84.73	57.14	80.48	74.43	88.63	82.70	88.13	94.25	96.63	81.68	91.34	54.07	76.74	78.03	90.05	65.55	84.52	83.09	74.58	52.15	84.09
Ours	**91.89**	**91.59**	88.41	**81.08**	90.65	90.01	**97.57**	**97.00**	**99.17**	**94.89**	**96.56**	74.65	88.37	80.63	**94.65**	80.96	97.22	**94.26**	**79.13**	**87.22**	94.11

TABLE XI

TOP-1 ACCURACY (%) RESULTS OF OUR METHOD AND SIX EXISTING METHODS ON THE TEST SET OF HRSC2016. THE AVERAGE ACCURACY AND ACCURACY OF EACH CLASS ARE LISTED, AND THE BEST ACCURACY IS BOLDED. AVE_ACC IS THE AVERAGE ACCURACY

		Head						Tail										
Method	Ave_acc	HR0	HR1	HR2	HR3	HR4	HR5	HR6	HR7	HR8	HR9	HR10	HR11	HR12	HR13	HR14	HR15	HR16	HR17
SADE [58]	73.12	53.89	77.42	65.91	73.40	86.49	**89.47**	60.98	**97.73**	**93.75**	60.87	66.67	91.67	**64.71**	**76.92**	**60.00**	50.00	**99.99**	**75.00**
TLC [35]	61.62	53.29	69.35	52.27	60.64	69.37	71.05	53.66	86.36	84.38	60.87	45.45	77.78	35.29	38.46	26.67	30.00	90.00	50.00
BKD [63]	46.35	65.27	35.48	45.45	55.32	38.74	47.37	21.95	86.36	43.75	34.78	18.18	36.11	5.88	30.77	40.00	30.00	80.00	37.50
RIDE [60]	73.13	53.29	78.23	73.86	78.72	82.88	**89.47**	**68.29**	90.91	**93.75**	**78.26**	60.61	83.33	58.82	69.23	46.67	60.00	**99.99**	50.00
ResLT [61]	66.37	51.50	77.42	62.50	60.64	74.77	86.84	43.90	95.45	81.25	73.91	63.64	66.66	47.06	69.23	33.33	40.00	**99.99**	**75.00**
LAL [62]	43.14	57.49	54.03	47.73	38.30	34.23	57.89	4.88	93.18	12.50	21.74	9.09	27.78	0.00	46.15	40.00	50.00	60.00	12.50
Ours	**78.98**	**69.46**	**88.71**	**85.23**	**90.43**	**89.19**	81.58	65.85	81.82	84.38	65.22	**69.70**	**94.44**	5.88	**76.92**	33.33	**70.00**	80.00	62.50

TABLE XII

TOP-1 ACCURACY (%) RESULTS OF OUR METHOD AND SIX EXISTING METHODS ON THE TEST SET OF XVIEW. AVE_ACC IS THE AVERAGE ACCURACY

Method	Ave_acc
SADE [58]	51.39
TLC [35]	60.77
BKD [63]	52.85
RIDE [60]	53.48
ResLT [61]	58.93
LAL [62]	56.57
Ours	**61.35**

ResNet-50 is adopted as our backbone. Our student network has 23.5M parameters, and takes about 6.3 ms to process a sample.

V. CONCLUSION

We present a CFCL for the long-tailed remote sensing object recognition. Head-tail center feature generation is first implemented to estimate the central feature of each class, which is achieved by using the head samples and tail samples to train two teacher models, respectively. That will avoid the tail class feature learning being affected by the head classes. Then, CFCL is proposed to make features of each class more compact, which is realized by distilling the central features to the student model. Since the central features are estimated by selecting the features correctly classified by teacher models, the inaccurate knowledge is abandoned. Thus, the student model can learn better feature representations and obtain clearer classification boundaries. Extensive experimental results show the superior performance of our proposed CFCL compared to existing state-of-the-art methods.

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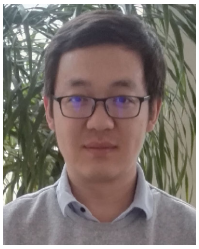
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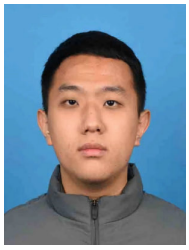
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